

PRIVACY-FIRST COMPLIANCE LAYER FOR REGULATED AUTOMATED SALES

Secure age verification for automated retail in regulated sectors

A proof-of-concept system that combines age decisioning, identity assurance and audit-safe traceability without storing unnecessary personal data.

51

automated tests

94%

test coverage

0 PII

in audit logs

The problem

Automated sales of restricted products require more than a normal vending flow. Operators must verify legal age, reduce fraud and keep evidence of responsible sale while avoiding unnecessary privacy risk.

The solution

A modular verification layer for kiosks and self-service terminals: document-reading interface, certificate-validation interface, liveness concept, 1:1 face match, legal-age calculation and HMAC-based audit token.

Initial use case

Licensed tobacco retail premises in Spain. The long-term opportunity is broader: regulated vending, alcohol, pharmacy/parapharmacy, identity-sensitive access and other age-restricted environments.

Business model

- Software licensing per terminal/device.
- SaaS compliance dashboard.
- Integration and pilot project fees.
- Maintenance and support contracts.
- Hardware partner/OEM model.

Verification flow

Document
interfaceSignature
validationLiveness + face
match

Age decision

Audit-safe result

Current assets

- Python verification workflow.

Next milestone

- Legal and RGPD feasibility review.
- 10–20 discovery interviews.

- Mockable interfaces for hardware-dependent components.
- Automated test suite and demo script.
- Touchscreen mockup and architecture documentation.
- Business validation package and contact map.

- Hardware/integration partner search.
- Pilot-ready MVP budget and roadmap.

Privacy by design**Regulated retail****Age verification****Auditability****Python + pytest****B2B SaaS potential**

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github.com/Ana-lopez-web/kiosk-biometric-verification